

# RevIN Experiment 3: Centering and Transform Variants

## 1 Theoretical overview and goal

Mean centering assumes the look-back average is the appropriate location anchor for the future. Last-value centering instead predicts changes around the most recent observation, which may better suit local trends. The inverse hyperbolic sine transform

$$z = \operatorname{asinh}(x)$$

is approximately linear near zero and logarithmic for large magnitudes, giving a reversible compression of heavy-tailed values before instance normalization.

**Experimental goal.** Test two lightweight changes to RevIN’s inductive bias: a more local location anchor and a nonlinear magnitude compression as appendix ablations.

## 2 Implementation details

The variants use non-affine instance normalization and normalized MSE:

| Method                          | Center           | Transform          |
|---------------------------------|------------------|--------------------|
| <code>instance_nmse</code>      | Look-back mean   | Identity           |
| <code>revin_last_nmse</code>    | Last observation | Identity           |
| <code>revin_arcsinh_nmse</code> | Look-back mean   | <code>asinh</code> |

Every transformation is inverted before data-space evaluation. These appendix variants run in the large profile only, over the same six datasets, seven settings, DLinear/PatchTST models, five seeds, optimizer, batch size, 10,000-step budget, and validation frequency. Configuration mapping is defined by `method_args` in `src/slurm/run_revin_exp`. Layer behavior is in `src/utils/models.py`.

Tables are separate by model and test split. They report ordinary MSE as  $\text{mean} \pm \text{sample standard deviation}$  with explicit  $\times 10^m$  row scaling. The baseline for both contrasts is `instance_nmse`.

## 3 Interpretation

Last centering tests sensitivity to local level evolution; `asinh` tests sensitivity to tails and multiplicative scale. A gain is evidence for that specific preprocessing bias, not a general solution to conditional shift. Results should be examined across datasets because either transform may remove useful absolute-level information.